Drive Wheel Torque Calculation

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The torque the drive motor must supply at the wheel is the wheel radius times the total force needed to overcome rolling resistance, climb the incline, and accelerate the load. A safety factor is then applied to get the design torque.

Given Data

Wheel radius	R	48 mm = 0.048 m
Rolling resistance coefficient (rubber on concrete)	C_f	0.02
Total weight of the loaded vehicle	W	1470 N
Mass of the loaded vehicle	m	150 kg
Acceleration	a	0.447 m/s^2
Incline angle (1:12 ramp)	θ_{ramp}	4.76°
Design incline angle (rounded up)	θ	5°
Safety factor	SF	1.5

Forces at the Wheel

Rolling resistance

$F_R = W \cdot C_f \cdot \cos\theta = (1470\ \text{N})(0.02)\cos(5°)$ **29.29 N**

Grade (incline) force

$F_B = W \cdot \sin\theta = (1470\ \text{N})\sin(5°)$ **128.12 N**

Acceleration force

$F_A = m \cdot a = (150\ \text{kg})(0.447\ \text{m/s}^2)$ **67.05 N**

Total force

$F_{total} = F_R + F_B + F_A = 29.29\ \text{N} + 128.12\ \text{N} + 67.05\ \text{N}$ **224.46 N**

Torque

Required wheel torque

$T = R \cdot F_{total} = (0.048\ \text{m})(224.46\ \text{N})$ **10.77 N·m**

Design torque with the safety factor applied

$T_{design} = SF \cdot T = (1.5)(10.77\ \text{N·m})$ **16.16 N·m**

Note: the incline is a 1:12 ramp (4.76°), rounded up to 5° for a conservative design. Using 4.76° instead would give $T_{design} \approx 15.7$ N·m.